MATCAP1

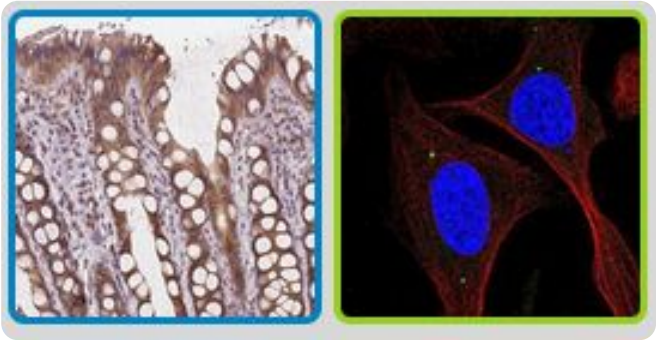
by Jas

Abstract

MATCAP1 (KIAA0895L), or microtubule associated tyrosine carboxypeptidase 1, is a gene that codes the protein by the same name, which controls microtubule dynamics and their functions. It may be able to be utilized to suppress the proliferation of cancer cells.

- NCBI
- GEO Profiles
- Human Protein Atlas
- STRING

colon, human cells

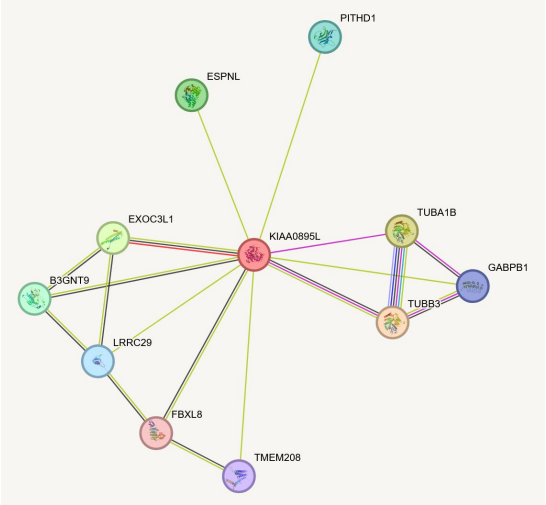


Intro

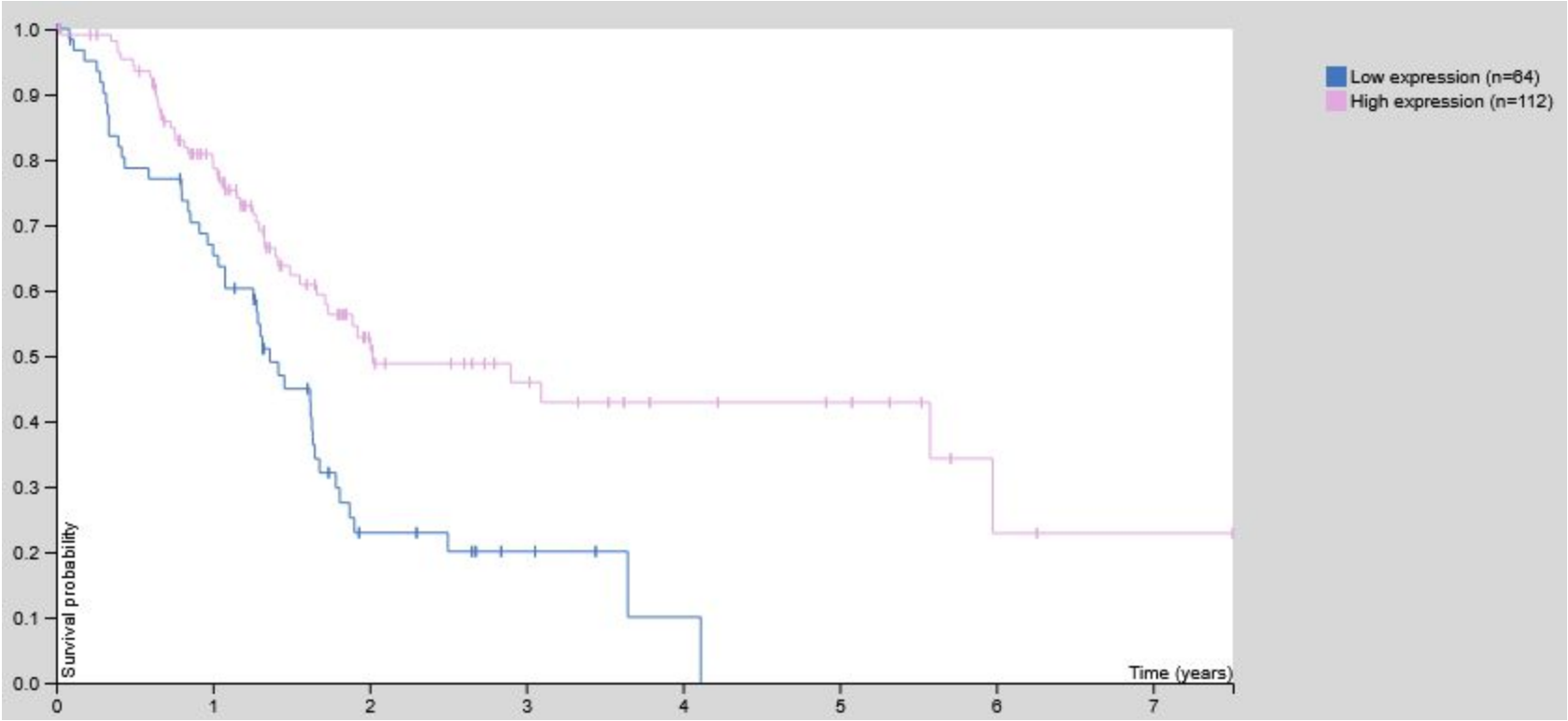
PROTEIN EXPRESSION AND LOCALIZATION	
Tissue profile[i]	General cytoplasmic expression.
Subcellular location[i]	Localized to the Nucleoplasm, Vesicles
Predicted location[i]	Intracellular

- microtubule associated tyrosine carboxypeptidase 1
- Known more frequently as KIAA0895L
- Codes protein by same name (KIAA0895L)
- Chromosome 16
- regulates microtubule dynamics and function
- Connected to many genes related to cytoskeleton formation and organization

e.g:
TUBA1B: formation of cytoskeleton
EXOC3L1: cytoskeleton organization



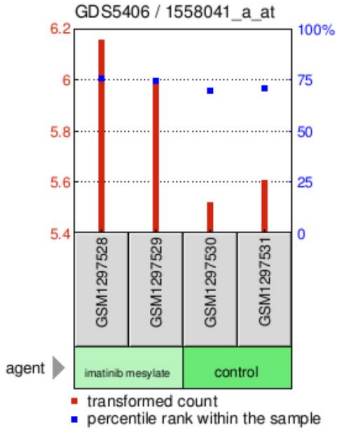
Pancreatic Cancer



Hypothesis #1:

MATCAP1 works to suppress pancreatic cancer cells

Profile GDS5406 / 1558041_a_at
Title Imatinib mesylate effect on EOL-1 eosinophilic leukemia cell line
Organism Homo sapiens

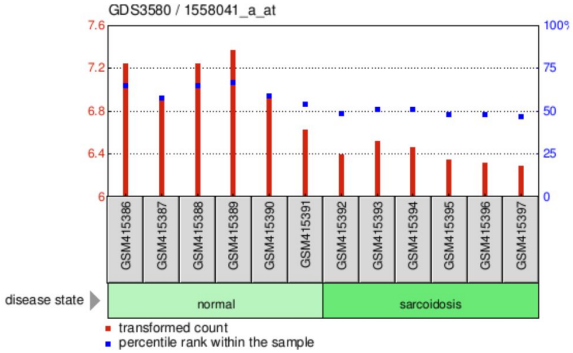


Graph caption help

Sample	Title	Value	Rank
GSM1297528	EOL1_IMA1	6.15869	76
GSM1297529	EOL1_IMA2	6.00562	75
GSM1297530	EOL1_CTRL3	5.52272	70
GSM1297531	EOL1_CTRL4	5.60794	71

higher expression of MATCAP when imatinib mesylate present
imatinib mesylate treats leukemia (cancer in blood cells)

Profile GDS3580 / 1558041_a_at
Title Pulmonary sarcoidosis
Organism Homo sapiens



Graph caption help

Sample	Title	Value	Rank
GSM415386	Lung-normal-rep1	7.24589	65
GSM415387	Lung-normal-rep2	6.89653	58
GSM415388	Lung-normal-rep3	7.24483	65
GSM415389	Lung-normal-rep4	7.37189	67
GSM415390	Lung-normal-rep5	6.93959	59
GSM415391	Lung-normal-rep6	6.62945	54
GSM415392	Lung-sarcoidosis-rep1	6.40261	49
GSM415393	Lung-sarcoidosis-rep2	6.52543	51
GSM415394	Lung-sarcoidosis-rep3	6.47223	51
GSM415395	Lung-sarcoidosis-rep4	6.35113	48
GSM415396	Lung-sarcoidosis-rep5	6.32186	48
GSM415397	Lung-sarcoidosis-rep6	6.29361	47

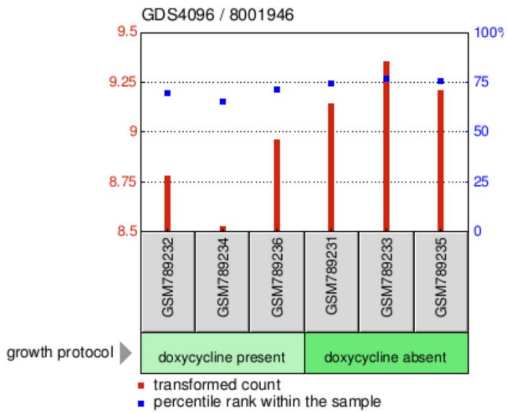
pulmonary sarcoidosis: small lumps of inflammatory cells in the lungs

lower expression of MATCAP1 when sarcoidosis is present

Hypothesis #2:

MATCAP1 is a tumor suppressor gene

Profile GDS4096 / 8001946
Title Mutant p53 depletion effect on MDA-468.shp53 breast tumor cell line
Organism Homo sapiens



Graph caption help

Sample	Title	Value	Rank
GSM789232	MDA-468.shp53 cells_3D culture, 8 days, +DOX_rep1	8.78368	70
GSM789234	MDA-468.shp53 cells_3D culture, 8 days, +DOX_rep2	8.51219	66
GSM789236	MDA-468.shp53 cells_3D culture, 8 days, +DOX_rep3	8.96595	72
GSM789231	MDA-468.shp53 cells_3D culture, 8 days, -DOX_rep1	9.14894	75
GSM789233	MDA-468.shp53 cells_3D culture, 8 days, -DOX_rep2	9.3579	77
GSM789235	MDA-468.shp53 cells_3D culture, 8 days, -DOX_rep3	9.21299	76

p53: encoded by tp53, tp53 is tsg

- As mutant, lose tumor suppressor properties

doxycycline: anticancer properties, able to control cancer progression through different mechanisms

https://www.sciencedirect.com/science/article/abs/pii/S0014299923004612

Microtubules

- Alters microtubule functions
- Mitosis, cell migration, cell signaling and trafficking
- If hijacked by cancer cells, would allow for proliferation
- MTAs already cancer therapy
- Control MATCAP1 to change function of microtubules to stop cancer

https://link.springer.com/article/10.1007/s00018-005-5330-x
https://www.sciencedirect.com/science/article/pii/S0171933520300145

Hypothesis #3:

MATCAP1 alters microtubule functions to prevent cancer proliferation

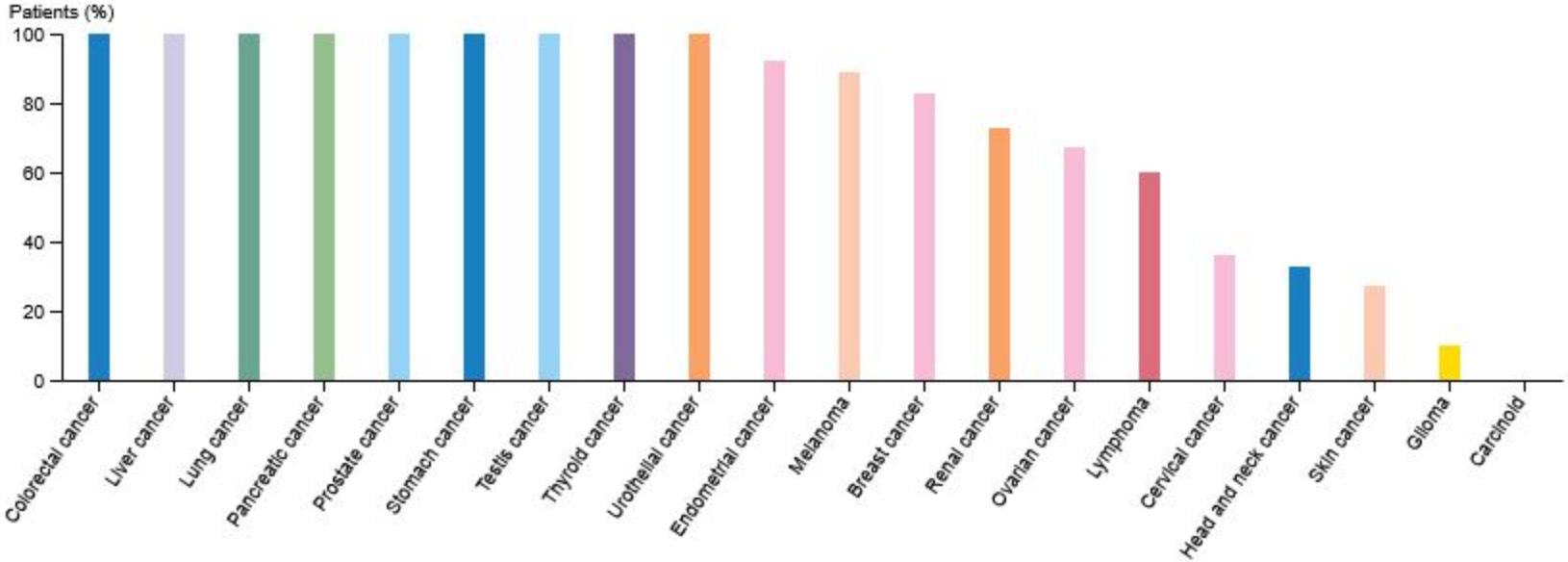
Conclusion

- Knockout mouse
- With doxycycline and without: what happens to MATCAP1 expression levels?

- See if MATCAP1 can be altered to suppress cancer (microtubules)
- Not only MTAs

Sources

- https://www.ncbi.nlm.nih.gov/
- https://www.proteinatlas.org/
- https://string-db.org/cgi/input?sessionId=bwEGgF6y3rKg&input_page_show_search=on
- https://scholar.google.com

Thank You!

Cancer tissues exhibited moderate to strong cytoplasmic immunoreactivity, with a granular pattern. Gliomas, carcinoids and several squamous cell carcinomas were weakly stained or negative.